

Handling Flash Deals with Soft Guarantee in Hybrid Cloud

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What are flash deals?



Apple Pre-Order



■ Amazon Prime Day

- Prime Day is a one-day-only global shopping event
- New deals are released as often as every five minutes

■ New iPhones pre-order

- iPhone 6 preorders were slated to start at midnight

■ WeChat red envelope

- WeChat has offered virtual red envelope containing virtual money that can be cashed out

Flash deals offer benefits to subscribers within short time!

envelope and hence the money

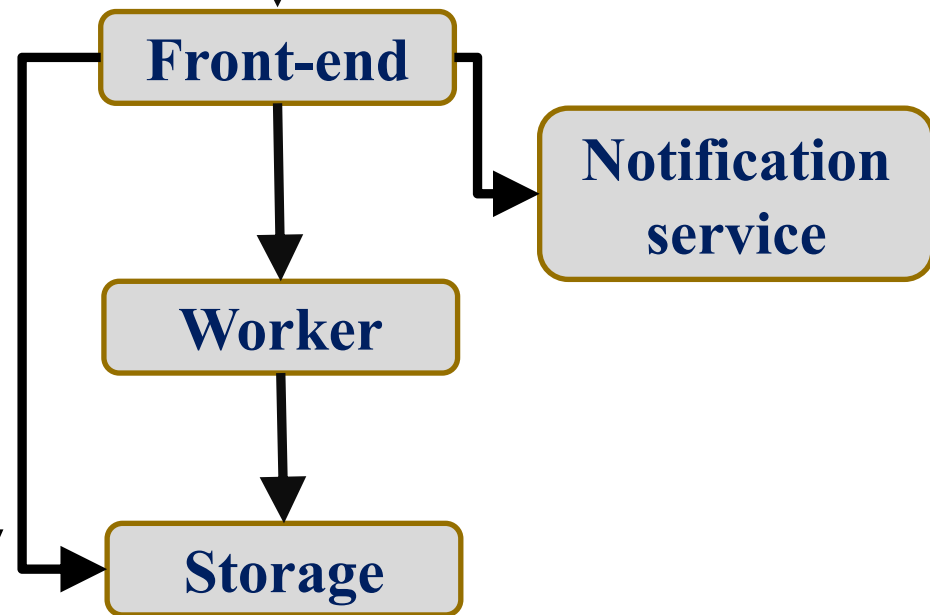
Fast & Simple

Simple

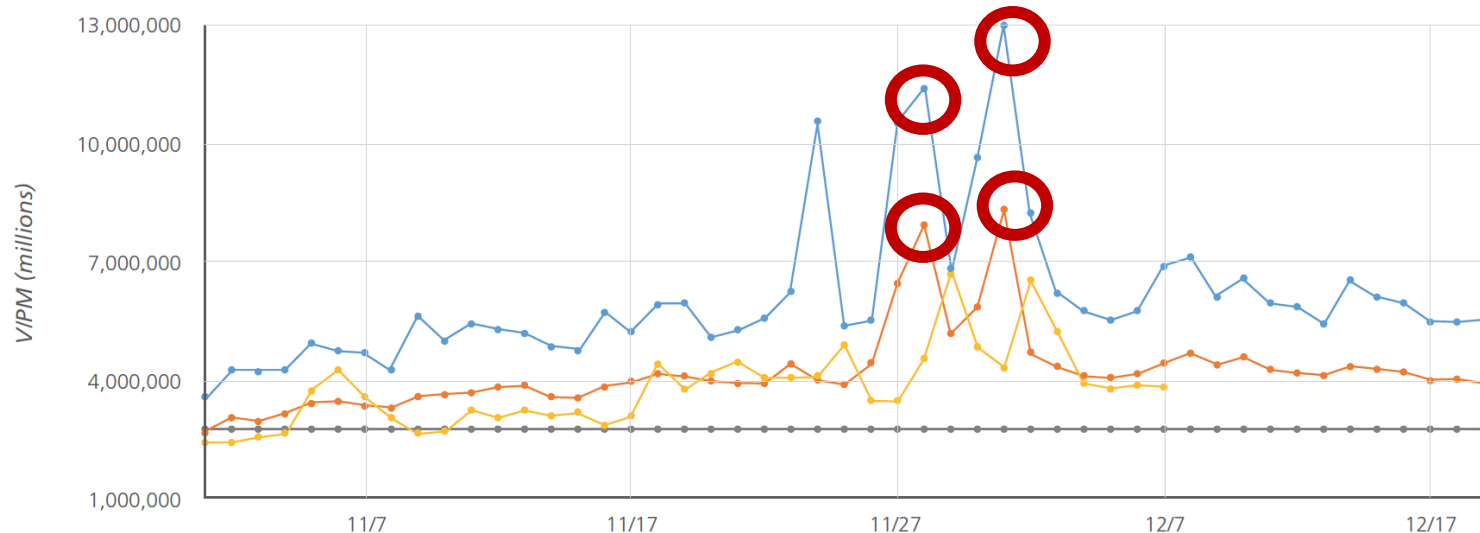
- Easy to get
 - ❑ One click on mouse
 - ❑ Shake smartphone
- Straightforward business logic
 - ❑ First some persons win

Fast

- Limited profit
 - ❑ Discounted merchandise
 - ❑ Newly released iPhone
 - ❑ WeChat Red Envelope
- Short duration
 - ❑ Refresh every 5 minutes
 - ❑ Midnight on release day
 - ❑ Spring Festival Gala



Yet crowded



- Sales on Amazon's Prime Day **exceeded** Black Friday in 2014
- The times of shaking phones reached a total of **11 billion** and a peak of **810 million per minute**
- The pre-orders exceeded **two million** in the first 24 hours, making Apple Store unresponsive

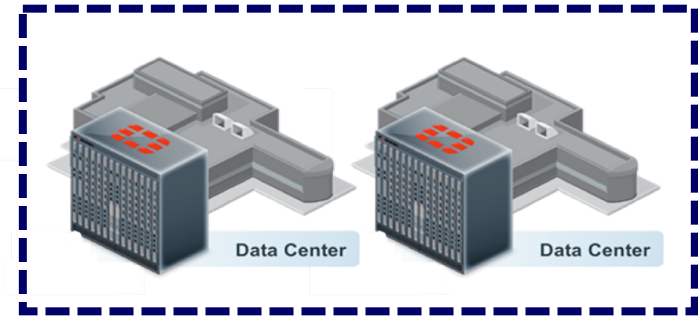
How to handle such fast, simple, and crowded flash deals?



Is private cloud OK?

■ Private cloud

- ❑ Dedicated datacenter or server cluster
- ❑ Virtual resources provided by cloud providers



Private cloud

■ Private cloud solution

❑ Advantages

- Enhanced security
- Ultimate control

❑ Disadvantages

- Limited capacity
- Low scalability
- Complex to operate

■ Requirement of security

- ❑ Protect confidential data

■ Requirement of performance

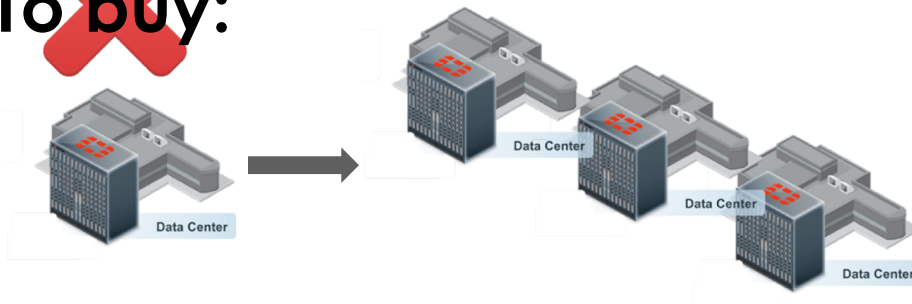
- ❑ Maximum uptime
- ❑ Fast page load time

How to increase capacity and improve scalability?



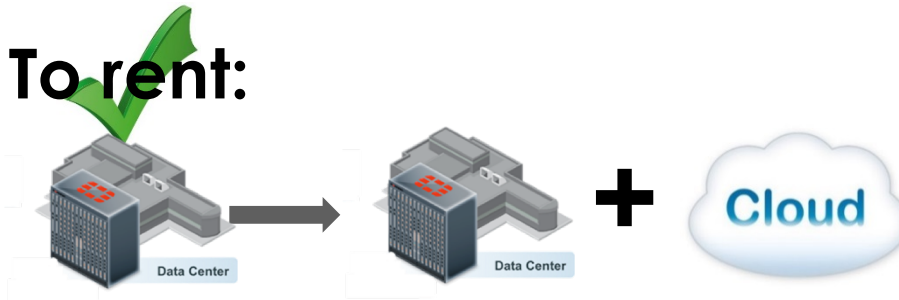
To buy or To rent?

To buy:



- Cost Increases linearly
 - Infrastructure
- Unable to scale up or down based on workloads
 - Temporary use

To rent:

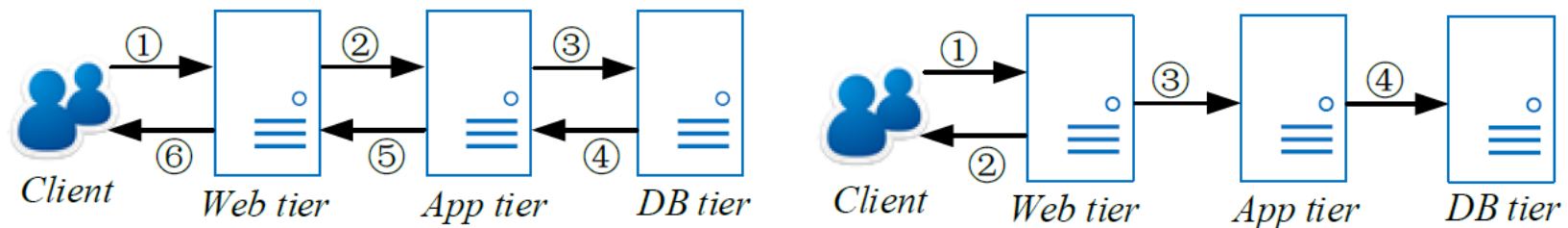


- Low price
- Auto scaling
 - Scalable capacity
 - Easy to operate
- Potentially unlimited resources

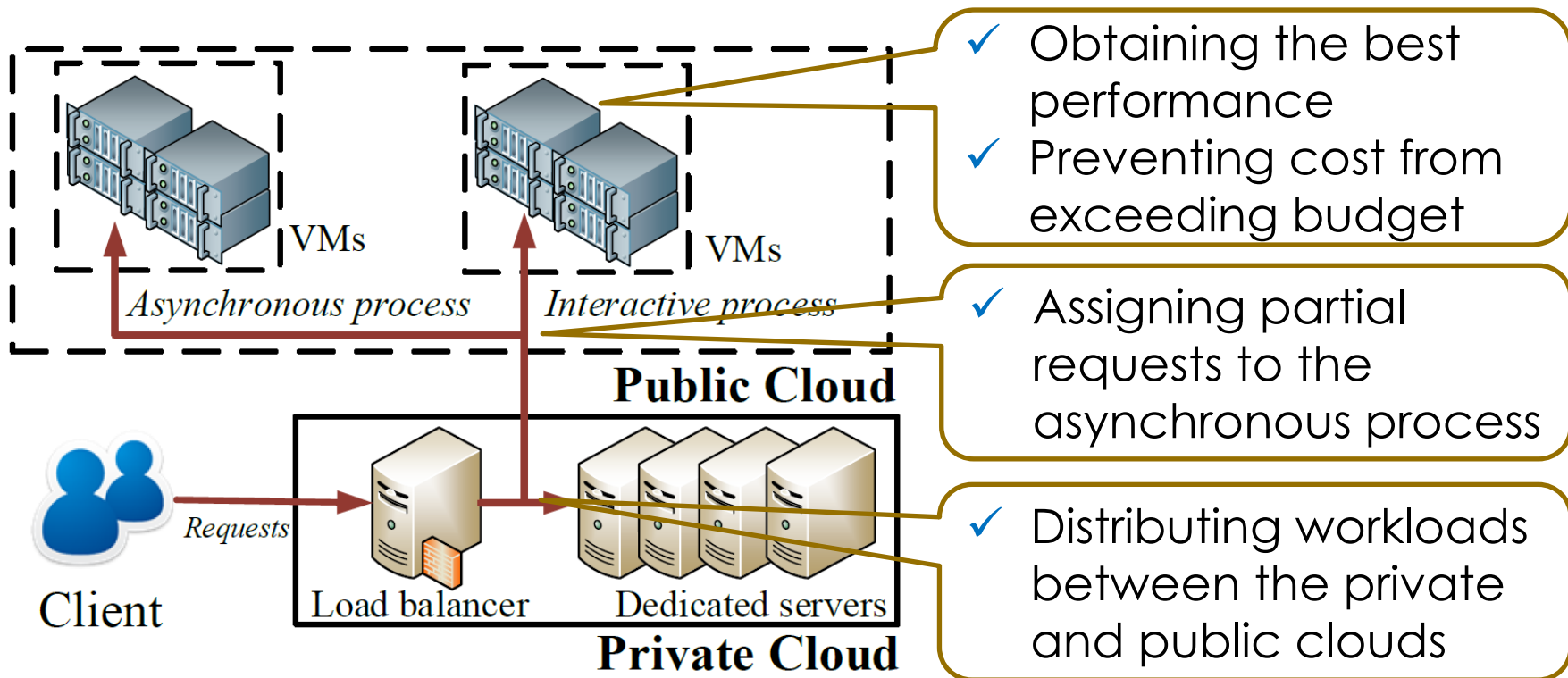
Hybrid cloud solution is a promising choice!

Is hybrid cloud enough?

- Revisit flash deals
 - Flash deals always bring benefits
 - Flash deals involve simple operations
- Postpone serving requests
 - Incentive to wait longer to get benefits
 - Serve partial requests instantly
 - Postpone serving others
- One example
 - Instead of waiting for the results returned from the application tier (1, 2, 3, 4, 5 in left)
 - Web servers send responses back to users (2 in right)
 - Guarantee the requests served asynchronously within deadline (3, 4 in right)



Hybrid cloud with soft guarantee



■ Problems

- ❑ Without prior knowledge of requests
- ❑ How to schedule requests
- ❑ How to adjust the scale of public cloud

Modeling flash deal applications

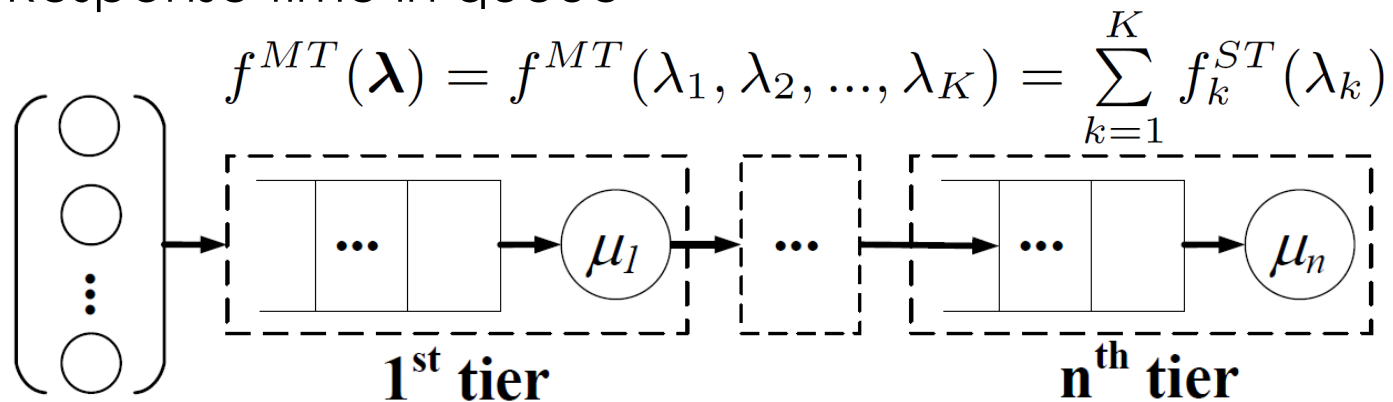
■ Single-tier Architecture [1][2][3][4]

- Request arrival follows Poisson process
- Service time is generally distributed
- Model the application as an M/G/1/PS queue
- Response time in queue

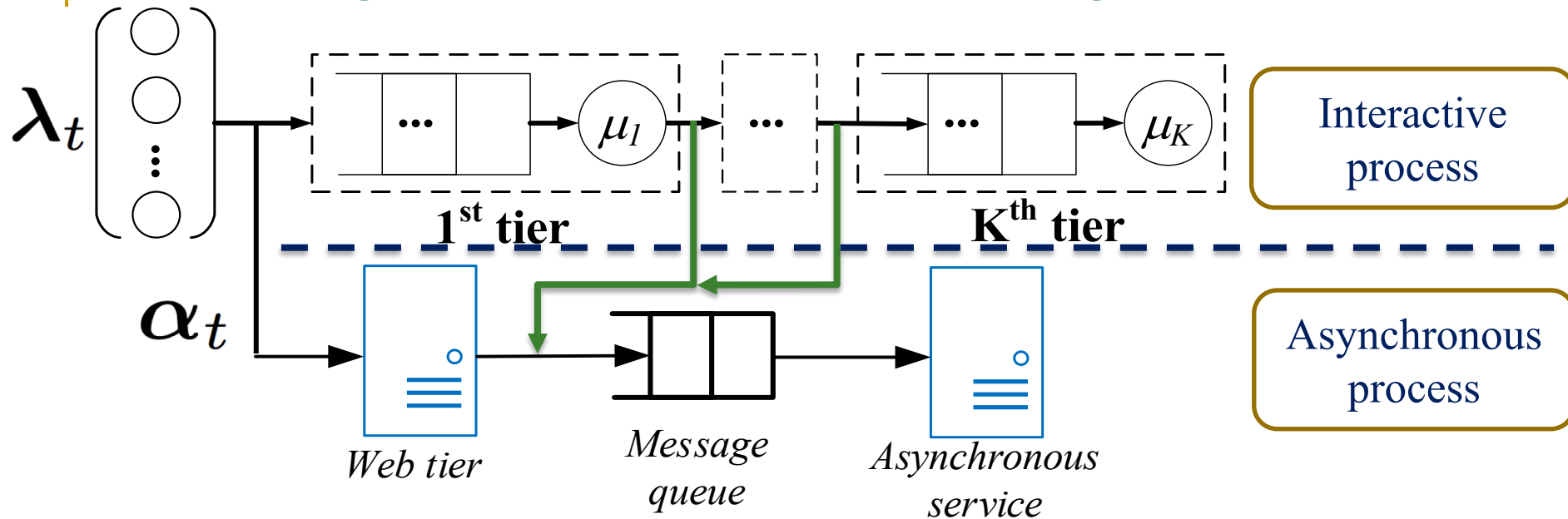
$$f^{ST}(\lambda) = \int_0^\infty \frac{x}{1-\rho} dF(x) = \frac{E[X]}{1-\rho}$$

■ Multi-tier Architecture [5][6][7]

- Lemma 1. the arrival rate $\lambda_{k+1} = \lambda_k$, when the queueing system is stable
- Response time in queue



Extending multi-tier with service degradation



■ Service degradation

- ❑ Each message binds to a series of tasks
- ❑ Classify messages into different priority classes $k \in \{1, 2, \dots, K\}$
- ❑ Model the asynchronous process as a priority queue

$$f^{AP}(\alpha) = \frac{C \sum_{k=1}^K \alpha_k}{(1 - \sum_{k=1}^{K-1} \frac{\alpha_k}{\mu^{AP}})(1 - \sum_{k=1}^K \frac{\alpha_k}{\mu^{AP}})}$$

Evaluating response time

- Private cloud – single tier
 - λ_t^V is the number of requests assigned to the private cloud during the t^{th} time slot
 - Model flash deals in private cloud as single-tier architecture
 - Response time can be evaluated as $d_t^V = f^{ST}(\lambda_t^V)$
- Public cloud – multi tier with soft guarantee
 - λ_t^U is the number of requests assigned to the public cloud during the t^{th} time slot
 - Model flash deals in private cloud as multi-tier architecture
 - Interactive process

$$d_t^{IP}(\boldsymbol{\lambda}_t, \boldsymbol{\alpha}_t) = f^{MT}(\boldsymbol{\lambda}_t - \boldsymbol{\alpha}_t)$$

- Asynchronous process

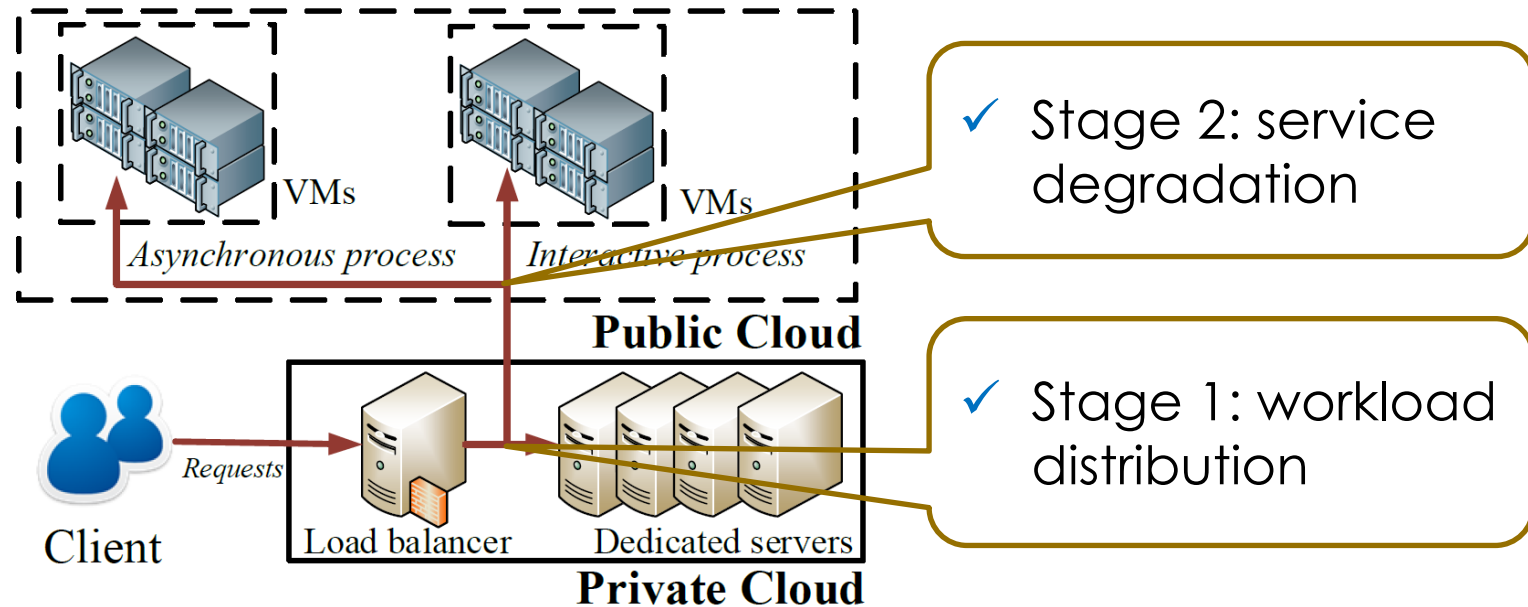
$$d_t^{SG}(\boldsymbol{\alpha}_t) = f^{AP}(\boldsymbol{\alpha}_t)$$

- Hybrid cloud

- Response time can be evaluated as

$$d_t^H = \frac{\lambda_t^V}{\lambda_t} d_t^V + \frac{\lambda_t^U}{\lambda_t} (d_t^{IP} + d_t^R) + \frac{\lambda_t - \lambda_t^V - \lambda_t^U}{\lambda_t} \cdot D$$

Request scheduling problem



- Workload distribution
 - Distributing requests between the private and public clouds

$$\begin{aligned} \min \quad & d_t^H(\lambda_t^V, \lambda_t^U) \\ \text{s.t.} \quad & \lambda_t^V + \lambda_t^U - \lambda_t \leq 0 \end{aligned}$$

- Service degradation
 - Assigning partial requests to asynchronous process

$$\begin{aligned} \min \quad & d_t^{IP}(\alpha_t) \\ \text{s.t.} \quad & d_t^{SG}(\alpha_t) \leq L \end{aligned}$$

Capacity adjusting problem

- We set a budget b
- The number of EC2 instances a tenant can boot is n
- The decision on leasing n EC2 instances is $\mathbf{x}_t^{(n)}$
- Performance-Cost ratio of leasing n EC2 instances

$$\pi_t^{(n)} = \frac{cap_t}{a_t \cdot n_t} = \frac{1}{a_t d_t^H \mathbf{J}^T \mathbf{x}_t^{(n)}}, \mathbf{J} = (1, 2, \dots, N)$$

- Problem formulation

$$\max \quad \sum_{t=1}^m \boldsymbol{\pi}_t^T \mathbf{x}_t$$

$$\text{s.t.} \quad \sum_{i=1}^t \mathbf{c}_i^T \mathbf{x}_i \leq b, \quad \mathbf{c}_i^T = a_i \cdot \mathbf{J}^T$$

$$i = 1, 2, \dots, m,$$

$$\boldsymbol{\pi}_t = (\pi_t^{(1)}, \pi_t^{(2)}, \dots, \pi_t^{(n)}, \dots, \pi_t^{(N)})$$

$$\mathbf{x}_t \in \{\mathbf{x}_t^{(1)}, \mathbf{x}_t^{(2)}, \dots, \mathbf{x}_t^{(n)}, \dots, \mathbf{x}_t^{(N)}\}$$

**Online
problem**

NP hard

Controlling cost

PC ratio

Capacity decision

Capacity adjusting algorithm

- Define a partial linear problem on $\{0, s\}$, where $s = \epsilon m$ and $0 < \epsilon < 1$

- The partial linear problem

$$\begin{aligned} \max \quad & \sum_{t=0}^s \pi_t^T \mathbf{x}_t \\ \text{s.t.} \quad & \sum_{t=0}^s \mathbf{c}_t^T \mathbf{x}_t \leq (1 - \epsilon)\epsilon b, \\ & \mathbf{x}_t \in K, t \in [0, s]. \end{aligned}$$

- The corresponding dual problem

$$\begin{aligned} \min \quad & (1 - \epsilon)\epsilon b \cdot p + \sum_{t=0}^s y_t \\ \text{s.t.} \quad & c_{tj} \cdot p + y_t \geq \pi_{tj}, j \in [1, N] \\ & p, y_t \geq 0, t \in [0, s]. \end{aligned}$$

- $(\hat{p}, \hat{\mathbf{y}})$ represents the optimal solution to problem

- Decision on capacity adjustment

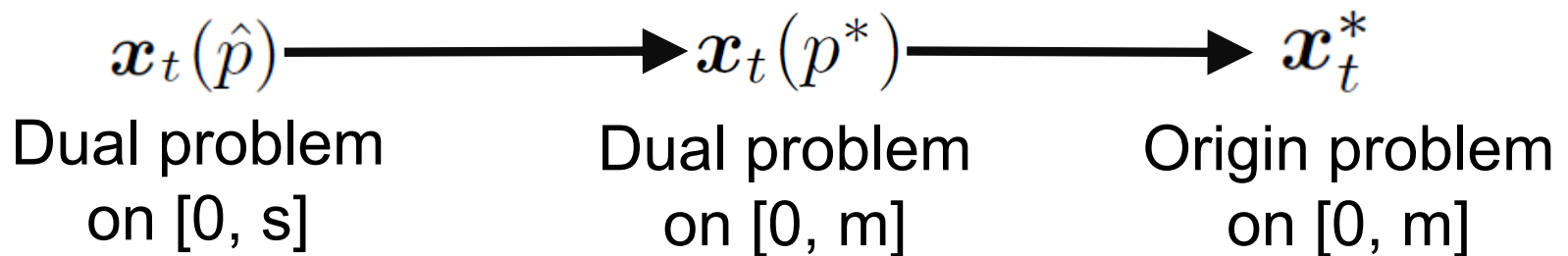
$$x_{tj}(p) = \begin{cases} 1, & \pi_{tj} > p \cdot c_{tj}, \\ 0, & \pi_{tj} < p \cdot c_{tj}. \end{cases} \quad x_{tj}(p) = \begin{cases} 1, & j = \arg \max_{j \in N} \{\pi_{tj} - p \cdot c_{tj}\}, \\ 0, & \text{else.} \end{cases}$$

Capacity adjusting algorithm

- Two special cases
 - If $\pi_{tj} = p \cdot c_{tj}$
 - If $\pi_{tj} - p \cdot c_{tj} = \pi_{tl} - p \cdot c_{tl}$
- The algorithm becomes ineffective
- Inspired by the existing literature [8][9], we make Assumption 1

Assumption 1. *For any p , there can be at most 1 column of $\pi_t \in \{\pi_i | i \in [1, m]\}$, i.e., π_{tj} , such that $\pi_{tj} = p \cdot c_{tj}$ or $\pi_{tj} - p \cdot c_{tj} = \pi_{tl} - p \cdot c_{tl}$.*

- Summary of capacity adjusting algorithm



Optimality analysis

- Q1: is $\mathbf{x}_t(p^*)$ the same to $\mathbf{x}_t^*$?

Lemma 2. For all $t \in [0, m]$, under Assumption 1, $\mathbf{x}_t(p^*)$ and $\mathbf{x}_t^*$ differs no more than 1 value of t .

- Q2: is $\mathbf{x}_t(\hat{p})$ accurate enough as a substitute to $\mathbf{x}_t(p^*)$?

Lemma 3. The primal solution derived using sample dual price $\hat{p}$ is a feasible solution to the linear problem with high probability of $1 - \epsilon$ that

$$\sum_{t=1}^m \mathbf{c}_t^T \mathbf{x}_t(\hat{p}) \leq b, \text{ given that } b \geq \frac{3m \ln(N/\epsilon)}{\epsilon^3}$$

- Q3: how much is the gap between $\mathbf{x}_t(\hat{p})$ and $\mathbf{x}_t^*$?

Lemma 4. The primal solution constructed using sample dual price $\hat{p}$ is a near-optimal solutions to the linear problem with high probability of $1 - \epsilon$ that

$$\sum_{t=1}^m \pi_t^T \mathbf{x}_t(\hat{p}) \geq (1 - 3\epsilon)OPT, \text{ given that } b \geq \frac{3m \ln(N/\epsilon)}{\epsilon^3}$$

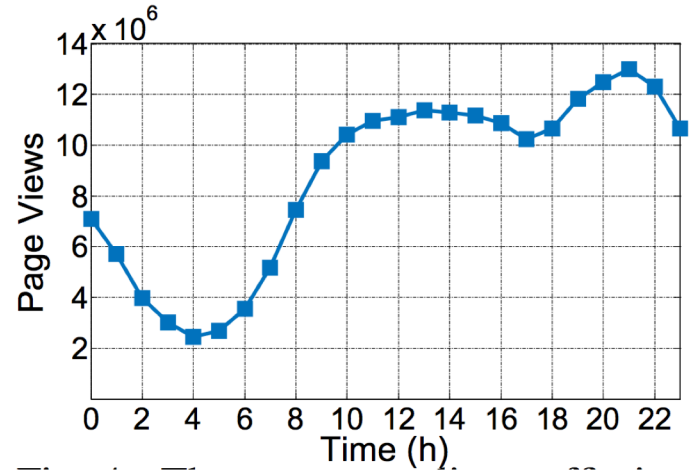
- Q4: how much is the gap between OPT and the algorithm?

Proposition 1. For any $\epsilon > 0$, the Capacity-Adjusting Algorithm is $1 - 6\epsilon$ competitive for the online linear problem

Evaluation

■ Real world trace

- ❑ Online traffic in U.S. on Cyber Monday measured by Akamai



■ Testbed

- ❑ A private cloud on two servers with OpenStack Mitaka
- ❑ A public cloud 20 EC2 large type instances on AWS

■ Implementation

- ❑ The web tier is deployed by an Apache HTTP server
- ❑ Two Tomcat 9.0 servers as the application tier,
- ❑ Use HttpClient 4.5.2 to generate requests
- ❑ A Servlet querying records of a table from a MySQL database

Evaluation

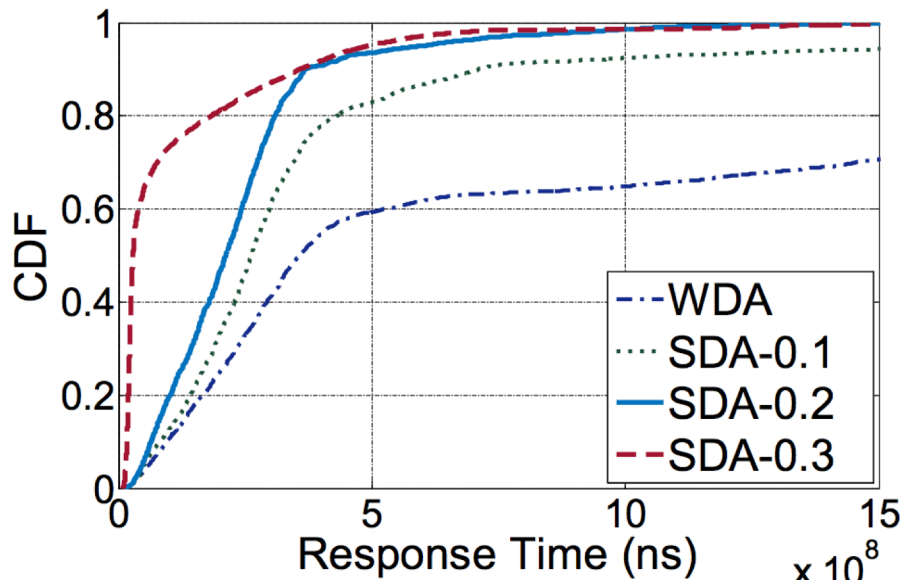


Fig. 5. CDF of response time of service degradation algorithms.

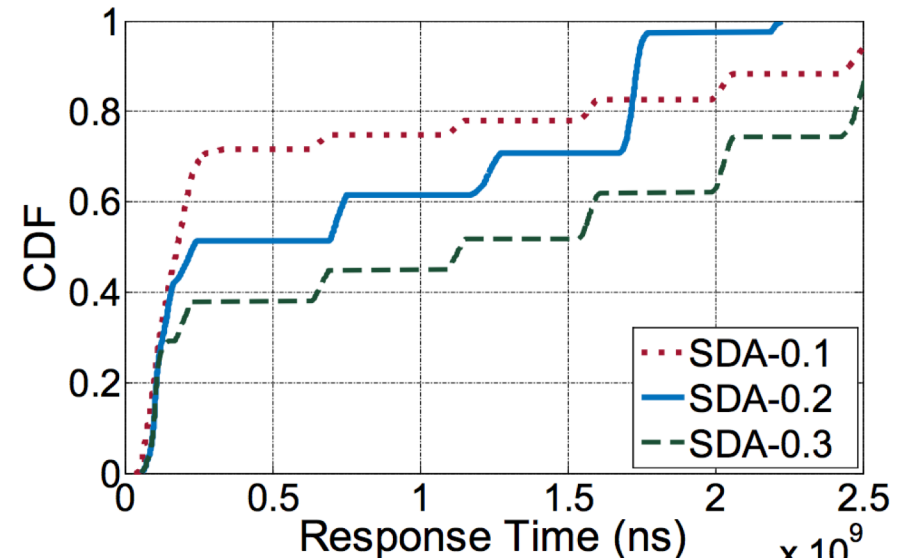


Fig. 6. CDF of execution time among postponed requests.

- Obviously, scheduling more requests to the asynchronous process can reduce response time remarkably
- The response time of the asynchronous process can be controlled within predefined deadline

Evaluation

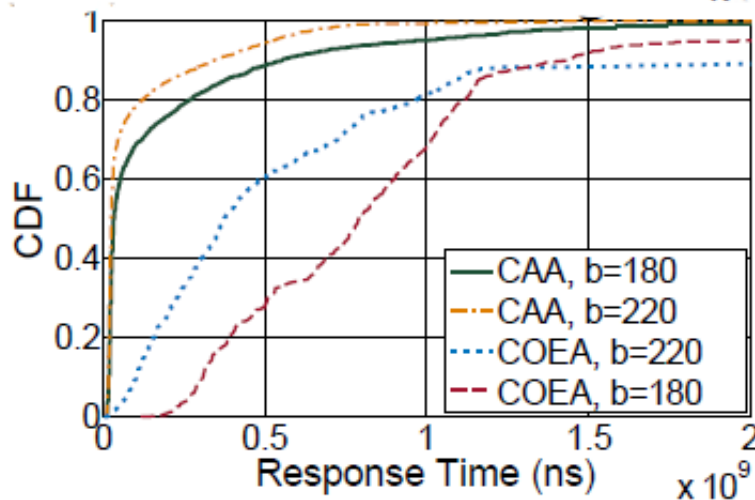


Fig. 12. CDF of response time of CAA and CEOA.

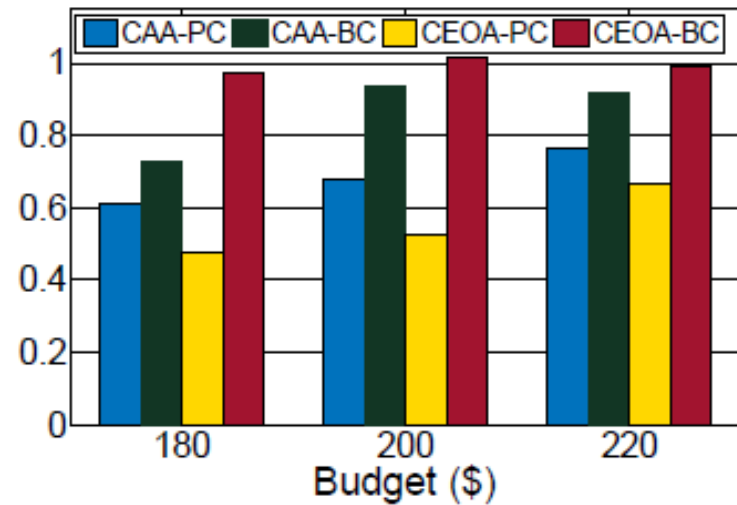


Fig. 13. PC and BC ratios of CAA and CEOA.

- Compared with CEOA, CAA reduces response time by 15% and improves the PC ratio by 19% on average, respectively

Conclusion

- We proposed a solution for flash deal applications to withstand flash crowds in a hybrid cloud
 - Concerning scheduling requests, we achieved fast response time of the interactive process as well as guaranteed requests served in the asynchronous process within a predefined deadline
 - In terms of adjusting capacity, we tuned scale of the public cloud with the objectives of performance-cost ratio maximization as well as outsourcing cost minimization.
 - Compared with previous work, our solution reduced response time by 15% on average and effectively maintained cost within the budget.
-

Reference

No.	Paper	Source
[1]	Modeling differentiated services of multi-tier web applications	MASCOTS06
[2]	Preserving qos of e- commerce sites through self-tuning: A performance model approach	EC '01
[3]	Autonomous resource provisioning for multi-service web applications	WWW'10
[4]	Provisioning Servers in the Application Tier for E-commerce Systems	IWQoS'04
[5]	Agile dynamic provisioning of multi-tier internet applications	TAAS
[6]	Con- trolling quality of service in multi-tier web applications	ICDCS'06
[7]	An analytical model for multi-tier internet services and its applications	SIGMETRICS'05
[8]	The adwords problem: Online keyword matching with budgeted bidders under random permutations	EC'05
[9]	A dynamic near-optimal algorithm for online linear programming	Operations research

Q&A

Thank You!

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